SEMESTER I

AI6101: APPLIED STATISTICS AND PROBABILITY

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Measure of central tendency

Measures of central tendency are statistical tools used to identify the centre or typical value in a data set.

The three main types are **mean**, **median**, and **mode**.

- **Mean** is the average of all values.
- **Median** is the middle value when data is arranged in order.
- **Mode** is the value that appears most frequently.

These measures help summarize and compare data distributions in a simple way.

Measure of central tendency: mean

Mean (Average) is a measure of central tendency that represents the sum of all values divided by the number of values.

Formula:

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

Example:
Consider the numbers: 10, 15, 20, 25, 30
Sum = 10 + 15 + 20 + 25 + 30 = 100
Number of values = 5
Mean = 100 ÷ 5 = **20**

So, the mean of the given data is **20**.

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

Measure of central tendency: median

Median is the middle value of a data set when the numbers are arranged in ascending or descending order. It divides the data into two equal halves.

Steps to find the Median:

- Arrange the data in order.
- If the number of values is **odd**, the median is the middle value.
- If the number of values is **even**, the median is the average of the two middle values.

Measure of central tendency: median

Example (odd count):

Data: 12, 15, 18, 22, 25

Median = **18** (middle value)

Example (even count):

Data: 10, 20, 30, 40

Median = (20 + 30) ÷ 2 = **25**

So, the median represents the central value of an ordered data set.

Measure of central tendency: mode

Mode is the value that appears **most frequently** in a data set. A set can have one mode (**unimodal**), more than one mode (**bimodal** or **multimodal**), or no mode at all.

Steps to find the Mode:

1. Count the frequency of each value.
2. The value with the highest frequency is the mode.

Measure of central tendency: mode

Example 1 (one mode):

Data: 4, 7, 7, 9, 10

Mode = **7** (occurs twice)

Example 2 (two modes):

Data: 3, 3, 6, 6, 9

Modes = **3 and 6** (both occur twice)

If no number repeats, the data set has **no mode**. Mode is useful for categorical data or to know the most common value.

Measures of dispersion- Range

Range is a simple measure of dispersion that shows the **spread** or **difference** between the highest and lowest values in a data set.

Formula:

Range=Maximum value−Minimum value

Example:

Data: 8, 12, 15, 20, 25

Maximum = 25, Minimum = 8

Range = 25 − 8 = **17**

The range indicates how widely the values in a set are spread out. However, it is sensitive to extreme values (outliers).

Quartile Deviation

$$\text{Quartile Deviation} = \frac{Q3 - Q1}{2}$$

Quartile Deviation, also known as **Semi-Interquartile Range**, measures the spread of the middle 50% of a data set. It is based on the first quartile (Q1) and third quartile (Q3).

Formula:

$$\text{Quartile Deviation} = \frac{Q3 - Q1}{2}$$

- **Q1** is the 25th percentile (lower quartile)
- **Q3** is the 75th percentile (upper quartile)

Quartile Deviation

Example:

If Q1 = 20 and Q3 = 40

$$\text{Quartile Deviation} = \frac{40 - 20}{2} = \frac{20}{2} = \mathbf{10}$$

Quartile deviation is less affected by extreme values and is useful for skewed distributions.

Mean Deviation

Mean Deviation is a measure of dispersion that shows the average of the absolute differences between each data value and the mean or median of the data set.

Formula:

$$\text{Mean Deviation} = \frac{\sum |x_i - \bar{x}|}{n}$$

Where:

- x_i = each value
- $\bar{x}$ = mean of the data
- n = number of values

Mean Deviation

Example:

Data: 4, 6, 8

Mean $= (4 + 6 + 8)/3 = 6$

Deviations: $|4-6| = 2$, $|6-6| = 0$, $|8-6| = 2$

Mean Deviation $= (2 + 0 + 2)/3 =$ **1.33**

Mean deviation gives a better sense of average variability but ignores the direction (positive or negative) of deviations.

Standard Deviation

Standard Deviation is a widely used measure of dispersion that shows how much the values in a data set **deviate from the mean** on average.

Formula:

$$\text{Standard Deviation}(\sigma) = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n}}$$

Where:

- x_i = each value
- $\bar{x}$ = mean of the data
- n = number of values

Standard Deviation

Example:

Data: 2, 4, 6

Mean $= (2 + 4 + 6)/3 = 4$

Deviations squared: $(2-4)^2 = 4$, $(4-4)^2 = 0$, $(6-4)^2 = 4$

Variance $= (4 + 0 + 4)/3 = 2.67$

Standard Deviation $= \sqrt{2.67} \approx$ **1.63**

Standard deviation gives a clear picture of how data is spread around the mean and is widely used in statistical analysis.

Coefficient of variance

Coefficient of Variation (CV) is a measure of **relative dispersion**, showing the ratio of the standard deviation to the mean. It helps compare the variability of two or more data sets with different units or means.

Formu

$$\text{Coefficient of Variation (CV)} = \left(\frac{\text{Standard Deviation}}{\text{Mean}}\right) \times 100$$

Coefficient of variance

Example:

If Mean = 50 and Standard Deviation = 5

$$CV = \left(\frac{5}{50}\right) \times 100 = 10\%$$

A **lower CV** indicates **less variability**, and a **higher CV** indicates **more variability** relative to the mean. CV is especially useful for comparing consistency between data sets.

Skewness,

Skewness:

It measures the **asymmetry** of a distribution.

- **Positive Skew (Right-skewed):** Tail is longer on the right; mean > median
- **Negative Skew (Left-skewed):** Tail is longer on the left; mean < median
- **Zero Skewness:** Data is symmetric (mean = median)

Example: Income data often shows **positive skewness**.

Skewness

Meaning: Skewness measures the asymmetry (tilt) of a probability distribution or dataset.

Interpretation:

- **Skewness = 0** → The data is perfectly symmetric (like a normal distribution).
- **Positive Skew (Right-skewed)** → The right tail is longer; most values are on the left. Example: income distribution.
- **Negative Skew (Left-skewed)** → The left tail is longer; most values are on the right. Example: age at death (in certain populations).

Formula (sample skewness):

$$Skewness = \frac{\frac{1}{n}\sum(x_i - \bar{x})^3}{\left(\frac{1}{n}\sum(x_i - \bar{x})^2\right)^{3/2}}$$

where x_i = data points, $\bar{x}$ = mean, n = sample size.

Kurtosis

Kurtosis:

- It measures the **"tailedness"** or **peakedness** of a distribution.
- **Leptokurtic (High Kurtosis):** Sharp peak, heavy tails
- **Platykurtic (Low Kurtosis):** Flat peak, light tails
- **Mesokurtic (Normal Kurtosis):** Bell-shaped, like a normal distribution

Example: Stock return data may have **high kurtosis** due to extreme values.

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Example: Stock return data may have **high kurtosis** due to extreme values.

Kurtosis

Meaning: Kurtosis measures the "tailedness" or peakedness of a distribution compared to the normal distribution.

Interpretation:

- **Kurtosis = 3 (Mesokurtic)** → Same as normal distribution.
- **Kurtosis > 3 (Leptokurtic)** → **H**eavier tails, sharper peak (more outliers likely).
- **Kurtosis < 3 (Platykurtic)** → Flatter peak, thinner tails (fewer outliers).

Formula (sample kurtosis):

$$Kurtosis = \frac{\frac{1}{n}\sum(x_i - \bar{x})^4}{\left(\frac{1}{n}\sum(x_i - \bar{x})^2\right)^2}$$

- Often we report Excess Kurtosis = Kurtosis – 3 so that normal distribution has 0.

Karl Pearson's Coefficient of Skewness

- It measures the **degree and direction of asymmetry** of a frequency distribution.

Formula

$$\text{Skewness (Sk)} = \frac{\text{Mean} - \text{Mode}}{\text{SD}}$$

- If **Mode is not known**, then Pearson suggested an **approximation**:

$$\text{Sk} = \frac{3(\text{Mean} - \text{Median})}{\text{SD}}$$

Karl Pearson's Coefficient of Skewness

Interpretation

- **Sk = 0** → Symmetrical distribution (like Normal distribution).
- **Sk > 0** → Positively skewed (long tail to the right, mean > median > mode).
- **Sk < 0** → Negatively skewed (long tail to the left, mean < median < mode).

Step-by-Step Calculation (General Approach)

1. Find the **Mean** of the distribution.
2. Find the **Median** or **Mode** (depending on which is available).
3. Find the **Standard Deviation (SD)**.
4. Apply the formula.
5. Interpret the result (positive, negative, or zero skewness).

Karl Pearson's coefficient of skewness uses **mean, median/mode, and standard deviation** to measure asymmetry in data.

Practice Numerical

1. Mean

The marks obtained by 10 students in a test are:
25, 28, 30, 32, 35, 40, 42, 45, 50, 55.
Find the mean marks.

2. Median

The following are the ages of 11 players of a cricket team:
18, 20, 21, 22, 23, 24, 25, 26, 27, 28, 30.
Find the median age.

Practice Numerical

3. Mode

The marks obtained by students are:
10, 15, 20, 20, 25, 25, 25, 30, 35, 40.
Find the mode.

4. Range

The daily wages (in ₹) of 7 workers are:
120, 150, 180, 200, 220, 250, 300.
Find the range.

Practice Numerical

5. Quartile Deviation (Q.D.)

The incomes of 9 families (in ₹ ’000) are:
12, 15, 18, 20, 22, 24, 28, 30, 35.
Find the Quartile Deviation.

6. Mean Deviation (M.D.)

The marks obtained by students are:
5, 10, 15, 20, 25.

Find the Mean Deviation about the Mean.

Practice Numerical

7. Standard Deviation (S.D.)

The marks obtained in statistics by 5 students are:
12, 15, 20, 22, 25.
Find the Standard Deviation.

8. Coefficient of Variation (C.V.)

The monthly incomes (in ₹) of 5 persons are:
2000, 2200, 2500, 2800, 3000.
Find the Coefficient of Variation.

Practice Numerical

9. Skewness (Karl Pearson's Method)

The mean = 60, median = 55, and standard deviation = 10.
Find the coefficient of skewness.

10. Kurtosis (based on β_2)

For a distribution, the fourth central moment = 500, and the square of the second central moment = 100^2 = 10,000.
Find the coefficient of kurtosis ($\beta_2 = \mu_4 / \mu_2^2$).

Practice Numerical Answers

1. Mean

Sum = 382 → Mean = 382/10 = 38.2

2. Median

Ordered (11 values), middle = 6th value = 24

3. Mode

Most frequent = 25

4. Range

Max – Min = 300 – 120 = **180**

Practice Numerical Answers

5. Quartile Deviation (Q.D.)
Median = 22;
Lower half: 12,15,18,20 ⇒ $Q1 = (15+18)/2 = 16.5$
Upper half: 24,28,30,35 ⇒ $Q3 = (28+30)/2 = 29$
Q.D. = $(Q3 - Q1)/2 = (29 - 16.5)/2 =$ **6.25** (in ₹ ’000)

6. Mean Deviation about Mean
Mean = 15; |deviations| = 10,5,0,5,10 → sum = 30
M.D. = 30/5 = **6**

Practice Numerical Answers

7. Standard Deviation *(population)*
Mean = 18.8; variance = 22.16 → SD = $\sqrt{22.16}$ = **4.71** (approx)
8. Coefficient of Variation (C.V.) *(population)*
Mean = 2500; SD ≈ 368.78
C.V. = (SD/Mean)×100% ≈ (368.78/2500)×100% = **14.75%**
9. Skewness (Karl Pearson, using mean–median)
Sk = 3(Mean − Median)/SD = 3(60 − 55)/10 = **1.5** (positively skewed)
10. Kurtosis (β_2)
$\beta_2 = \mu_4/\mu_2^2$ = 500/10,000 = **0.05** (very platykurtic)

Questions

- Define **mean, median, and mode** with simple examples.
- Find the **mean** of: 10, 20, 30, 40, 50.
- Find the **range** of daily wages: 120, 150, 180, 200, 250.
- State the difference between **Skewness and Kurtosis**.
- Define **Coefficient of Variation**. Why is it used?
- The ages of 9 students are: 12, 13, 14, 15, 15, 16, 17, 18, 20.
  Find the median.
- Find the mode of: 5, 10, 10, 15, 20, 20, 20, 25, 30.
- The incomes of 5 families (in ₹'000) are: 20, 22, 25, 28, 30.
  Find the Coefficient of Variation.
- Define Quartile Deviation and illustrate with a numerical example.
- Explain positive and negative skewness with a diagram.

Questions

- The marks of students are: 12, 15, 20, 22, 25.

  Find the Mean Deviation about Mean.
- Calculate the Standard Deviation of: 2, 4, 6, 8, 10.
- For a distribution, Mean = 50, Median = 45, S.D. = 10.

  Find Karl Pearson's coefficient of Skewness.
- Write short notes on:

  a) Range and its limitations

  b) Importance of Standard Deviation

Questions

- The marks obtained by 10 students are:
  25, 28, 30, 32, 35, 40, 42, 45, 50, 55.
  Find the **Mean, Median and Mode**.
- The daily wages of 7 workers are:
  120, 150, 180, 200, 220, 250, 300.
  Calculate **Range, Quartile Deviation, and Standard Deviation**.
  ANS := Final Results
  Range = 180, Quartile Deviation = 50, Standard Deviation ≈ 56.26
- The mean of a distribution is 60, Median = 55, and S.D. = 10.
  Find **Skewness** and comment on the nature of distribution.
- For a distribution, the second central moment (μ_2) = 25, and the fourth central moment (μ_4) = 450.
  Find the **Kurtosis** and state whether the distribution is
  leptokurtic, mesokurtic, or platykurtic. Ans: